

Unit-6 Matrices



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$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \dots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix}$$

Matrix



Method:5 ➡ Eigen Value and Eigen Vector



What is Characteristic Equation?

→ Let $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ be a square matrix, $I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ be

an identity matrix and λ be a scalar .

→ $|A - \lambda I|$ is called **characteristic polynomial**.

→ The **characteristic equation** of the matrix A is $|A - \lambda I| = 0$.

How to find Characteristic Equation for 2×2 matrix?

1st Method

Example:1

Let $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, find characteristic equation of A.

Solution:

The characteristic equation of the matrix A is $|A - \lambda I| = 0$.

$$\rightarrow \text{Now, } A - \lambda I = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$\therefore A - \lambda I = \begin{bmatrix} -\lambda & 1 \\ 1 & -\lambda \end{bmatrix}$$

$$\rightarrow \text{Now, } |A - \lambda I| = 0$$

$$\Rightarrow \begin{vmatrix} -\lambda & 1 \\ 1 & -\lambda \end{vmatrix} = 0$$

$$\Rightarrow \lambda^2 - 1 = 0$$

Characteristic Equation

How to find Characteristic Equation for 2×2 matrix?

2nd Method

Example:1

Let $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, find characteristic equation of A.

Solution:

The characteristic equation of the matrix A is

$$\lambda^2 - S_1\lambda + S_2 = 0 \quad \text{.....} \rightarrow (1)$$

$$\begin{aligned} \text{where, } S_1 &= \text{trace}(A) = \text{Sum of principal diagonal entries of } A \\ &= 0 + 0 = 0 \end{aligned}$$

$$\rightarrow S_2 = |A| = -1$$

Now, put the value of S_1 and S_2 in equation (1)

$$\Rightarrow \lambda^2 - 0 \cdot \lambda - 1 = 0$$

$$\Rightarrow \lambda^2 - 1 = 0 \quad \rightarrow \text{Characteristic Equation}$$

How to find Characteristic Equation for 3×3 matrix?

1st Method

Example:2

Find the characteristic equation of the matrix $A = \begin{bmatrix} 2 & 1 & 2 \\ 0 & 2 & -1 \\ 0 & 1 & 0 \end{bmatrix}$

Solution:

The characteristic equation of the matrix A is $|A - \lambda I| = 0$

$$\rightarrow \text{Now, } A - \lambda I = \begin{bmatrix} 2 & 1 & 2 \\ 0 & 2 & -1 \\ 0 & 1 & 0 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 1 & 2 \\ 0 & 2 & -1 \\ 0 & 1 & 0 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

How to find Characteristic Equation for 3×3 matrix?

1st Method

Example:2

Find the characteristic equation of the matrix $A = \begin{bmatrix} 2 & 1 & 2 \\ 0 & 2 & -1 \\ 0 & 1 & 0 \end{bmatrix}$

Solution continue:

$$= \begin{bmatrix} 2 & 1 & 2 \\ 0 & 2 & -1 \\ 0 & 1 & 0 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$\therefore A - \lambda I = \begin{bmatrix} 2 - \lambda & 1 & 2 \\ 0 & 2 - \lambda & -1 \\ 0 & 1 & 0 - \lambda \end{bmatrix}$$

Now, $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} 2 - \lambda & 1 & 2 \\ 0 & 2 - \lambda & -1 \\ 0 & 1 & 0 - \lambda \end{vmatrix} = 0$$

$$\Rightarrow \lambda^3 - 4\lambda^2 + 5\lambda - 2 = 0$$

Characteristic Equation

How to find Characteristic Equation for 3×3 matrix?

2nd Method

Example:2

Find the characteristic equation of the matrix $A =$

$$A = \begin{bmatrix} 2 & 1 & 2 \\ 0 & 2 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

Solution:

The characteristic equation of the matrix A is

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0 \text{ (1)}$$

where, $S_1 = \text{trace}(A)$

$$= 2 + 2 + 0 = 4$$

→ $S_2 = \text{Sum of minors of principal diagonal elements}$

$$\therefore S_2 = \begin{vmatrix} 2 & -1 \\ 1 & 0 \end{vmatrix} + \begin{vmatrix} 2 & 2 \\ 0 & 0 \end{vmatrix} + \begin{vmatrix} 2 & 1 \\ 0 & 2 \end{vmatrix}$$

$$\therefore S_2 = 1 + 0 + 4 = 5$$

How to find Characteristic Equation for 3×3 matrix?

2nd Method

Example:2

Find the characteristic equation of the matrix $A = \begin{bmatrix} 2 & 1 & 2 \\ 0 & 2 & -1 \\ 0 & 1 & 0 \end{bmatrix}$

Solution continue: $S_3 = |A| = 2$

➔ Put the value of S_1, S_2 and S_3 in equation (1)

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

$$\Rightarrow \lambda^3 - 4\lambda^2 + 5\lambda - 2 = 0$$

Characteristic Equation

$$S_1 = 4, S_2 = 5, S_3 = 2$$

How to solve the **cubic** equation: $Ax^3 + Bx^2 + Cx + D = 0$

Example: Find the roots of the cubic equation $x^3 - 2x^2 - 15x + 36 = 0$.

Solution: **Step:1** Check for the values $x = -1, 1, -2, 2, -3, 3$

➡ Put $x = -1$ in the given equation

$$(-1)^3 - 2(-1)^2 - 15(-1) + 36 = -1 - 2 + 15 + 36 = 48 \neq 0$$

➡ Put $x = 1$ in the given equation

$$(1)^3 - 2(1)^2 - 15(1) + 36 = 1 - 2 - 15 + 36 = 20 \neq 0$$

➡ Put $x = -2$ in the given equation

$$(-2)^3 - 2(-2)^2 - 15(-2) + 36 = -8 - 8 + 30 + 36 = 50 \neq 0$$

How to solve the **cubic equation**: $Ax^3 + Bx^2 + Cx + D = 0$

Example: Find the roots of the cubic equation $x^3 - 2x^2 - 15x + 36 = 0$.

Solution continue:

➡ Put $x = 2$ in the given equation

$$(2)^3 - 2(2)^2 - 15(2) + 36 = 8 - 8 - 30 + 36 = 6 \neq 0$$

➡ Put $x = -3$ in the given equation

$$(-3)^3 - 2(-3)^2 - 15(-3) + 36 = -27 - 18 + 45 + 36 = 36 \neq 0$$

➡ Put $x = 3$ in the given equation

$$(3)^3 - 2(3)^2 - 15(3) + 36 = 27 - 18 - 45 + 36 = 0$$

$\therefore x = 3$ is one of the root of the given equation.



How to solve the **cubic** equation: $Ax^3 + Bx^2 + Cx + D = 0$

Example: Find the roots of the cubic equation $x^3 - 2x^2 - 15x + 36 = 0$.

Solution continue:

Step:2 Synthetic Division

$$\begin{array}{r|rrrr}
 3 & 1 & -2 & -15 & 36 \\
 & & 3 & 1 & -12 \\
 \hline
 & 1 & 1 & -12 & 0
 \end{array}$$

$3 \times 1 = 3$

$$x^3 - 2x^2 - 15x + 36 = 0$$

$$\therefore (x - 3)(x^2 + x - 12) = 0$$

$$\therefore (x - 3)(x - 3)(x + 4) = 0$$

$$\therefore (x - 3) = 0 \text{ or } (x - 3) = 0 \text{ or } (x + 4) = 0$$

$$\therefore x = 3, 3, -4$$

Multiplication of coefficient of 1st term and last term

Coefficient of middle term

Procedure to find the Eigen Value, Eigen Vector for 3×3 matrix

- 1) Write the Characteristic Equation $\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$ and find S_1, S_2 and S_3 .
- 2) Find the roots of $\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$, say $\lambda_1, \lambda_2, \lambda_3$.
Eigen values = $\lambda_1, \lambda_2, \lambda_3$.
- 3) Solve the system $[A - \lambda_1 I] \cdot \mathbf{X} = \mathbf{0}$ using Gauss Elimination Method.
- 4) Eigen vector = The solution $\mathbf{X}$ of above system.
- 5) Repeat step:4 for λ_2 & λ_3 to find remaining eigen vectors.

Method:5 Eigen values and Eigen vectors

Example:1

Find the eigen values and corresponding eigen vectors of the matrix

$$\begin{bmatrix} 5 & 0 & 1 \\ 1 & 1 & 0 \\ -7 & 1 & 0 \end{bmatrix}$$

Solution:

The characteristic equation of the matrix A is given by

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0 \text{} \textcolor{red}{(1)}$$

where, $S_1 = \text{trace}(A)$

$$= 5 + 1 + 0 = 6$$

 $S_2 =$ Sum of minors of principal diagonal elements

$$\therefore S_2 = \begin{vmatrix} 1 & 0 \\ 1 & 0 \end{vmatrix} + \begin{vmatrix} 5 & 1 \\ -7 & 0 \end{vmatrix} + \begin{vmatrix} 5 & 0 \\ 1 & 1 \end{vmatrix}$$

Method:5 $\Rightarrow$ Example:1(Continue)

$$\therefore S_2 = \begin{vmatrix} 1 & 0 \\ 1 & 0 \end{vmatrix} + \begin{vmatrix} 5 & 1 \\ -7 & 0 \end{vmatrix} + \begin{vmatrix} 5 & 0 \\ 1 & 1 \end{vmatrix}$$

$$\therefore S_2 = 0 + 7 + 5 = 12$$

$$\rightarrow S_3 = |A| = 5(0 - 0) - 0 + 1(1 + 7) = 8$$

$\rightarrow$ Put the value of S_1, S_2 and S_3 in equation (1)

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

$$\Rightarrow \lambda^3 - 6\lambda^2 + 12\lambda - 8 = 0 \cdots \cdots \cdots (2)$$

$\rightarrow$ Put $\lambda = 2$ in equation (2)

$$(2)^3 - 6(2)^2 + 12(2) - 8 = 8 - 24 - 24 + 8 = 0$$

$\therefore \lambda = 2$ is one of the root of the equation (2).

$$S_1 = 6, S_2 = 12 \\ \text{and } S_3 = 8$$

Method:5 $\Rightarrow$ Example:1(Continue)

$$\lambda^3 - 6\lambda^2 + 12\lambda - 8 = 0$$

$$\therefore (\lambda - 2)(\lambda^2 - 4\lambda + 4) = 0$$

$$\therefore (\lambda - 2)(\lambda - 2)(\lambda - 2) = 0$$

$$\therefore (\lambda - 2) = 0 \text{ or } (\lambda - 2) = 0 \text{ or } (\lambda - 2) = 0$$

$$\therefore \lambda = 2, 2, 2$$

Eigen Values

Handwritten solution for the cubic equation $\lambda^3 - 6\lambda^2 + 12\lambda - 8 = 0$. The polynomial is written in a grid. A red arrow points to the constant term -8, and another red arrow points to the coefficient of the linear term 12, with the text $2 \times 1 = 2$ below it. The synthetic division process is shown, resulting in the quotient $\lambda^2 - 4\lambda + 4$, which is then factored into $(\lambda - 2)(\lambda - 2)$.

Method:5 $\Rightarrow$ Example:1(Continue)

$\Rightarrow$ For $\lambda = 2$, the eigen vector is $(A - \lambda I)X = 0$

$$\Rightarrow (A - 2I)X = 0$$

$$\text{where, } B = A - 2I = \begin{bmatrix} 5-2 & 0 & 1 \\ 1 & 1-2 & 0 \\ -7 & 1 & 0-2 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 1 \\ 1 & -1 & 0 \\ -7 & 1 & -2 \end{bmatrix}$$

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } 0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 5 & 0 & 1 \\ 1 & 1 & 0 \\ -7 & 1 & 0 \end{bmatrix}$$

$$\Rightarrow \text{Consider, } [B \mid 0] = \left[\begin{array}{ccc|c} 3 & 0 & 1 & 0 \\ 1 & -1 & 0 & 0 \\ -7 & 1 & -2 & 0 \end{array} \right]$$

Method:5 $\Rightarrow$ Example:1(Continue)

$\rightarrow$ Consider,
$$\left[\begin{array}{ccc|c} 3 & 0 & 1 & 0 \\ 1 & -1 & 0 & 0 \\ -7 & 1 & -2 & 0 \end{array} \right]$$

Applying R_{12}

$$\sim \left[\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 3 & 0 & 1 & 0 \\ -7 & 1 & -2 & 0 \end{array} \right]$$

Applying $R_{12}(-3), R_{13}(7)$

$$\sim \left[\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 0 & 3 & 1 & 0 \\ 0 & -6 & -2 & 0 \end{array} \right]$$

Method:5 $\Rightarrow$ Example:1(Continue)

$$\begin{array}{c}
 \text{Applying } R_{12}(-3), R_{13}(7) \\
 \sim \left[\begin{array}{ccc|ccc} 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 3 & 1 & 0 & 0 & 0 \\ 0 & -6 & -2 & 0 & 0 & 0 \end{array} \right] \\
 \\
 \text{Applying } \left(\frac{1}{3}\right) R_2 \\
 \sim \left[\begin{array}{ccc|ccc} 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1/3 & 0 & 0 & 0 \\ 0 & -6 & -2 & 0 & 0 & 0 \end{array} \right] \\
 \\
 \text{Applying } R_{23}(6) \\
 \sim \left[\begin{array}{ccc|ccc} 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1/3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]
 \end{array}$$

Method:5 $\Rightarrow$ Example:1(Continue)

Applying $R_{23}(6)$

$$\begin{array}{c|ccc} & x & y & z \\ \hline 1 & 1 & -1 & 0 \\ 0 & 1 & 1 & 1/3 \\ 0 & 0 & 0 & 0 \end{array}$$

Here, $\rho(B) = 2$ & $\rho([B | 0]) = 3; z \in \mathbb{R}$

$\rho(B) = \rho([B | 0])$ Applying the Back Substitution,

where, $n = \text{number of variables}$, $z = 0$

Thus, the given system has **infinitely many** solution.

$$\Rightarrow y = -\frac{1}{3} \cdot z$$

$$\Rightarrow y = -\frac{1}{3}$$

$$\Rightarrow x - y = 0$$

$$\Rightarrow x = y$$

$$\Rightarrow x = -\frac{1}{3}$$

Method:5 $\Rightarrow$ Example:1(Continue)

$\Rightarrow$ Thus, the eigen vector corresponding to the eigen value $\lambda = 2$ is

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} t \\ -\frac{t}{3} \\ t \end{bmatrix}; t \in \mathbb{R}$$
$$= \begin{bmatrix} 1 \\ -\frac{1}{3} \\ 1 \end{bmatrix} \cdot t; t \in \mathbb{R}$$

$$\therefore X = \begin{bmatrix} 1 \\ -\frac{1}{3} \\ 1 \end{bmatrix}$$



Eigen Vector

Method:5 ➡ Eigen values and Eigen vectors

Example:2

Find the eigen values and corresponding eigen vectors of the matrix

$$\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}.$$

Solution:

The characteristic equation of the matrix A is given by

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0 \text{} \rightarrow (1)$$

where, $S_1 = \text{trace}(A)$

$$= 0 + 0 + 0 = 0$$

➡ $S_2 =$ Sum of minors of principal diagonal elements

$$\therefore S_2 = \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} + \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} + \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix}$$

Method:5 $\Rightarrow$ Example:2(Continue)

$$\therefore S_2 = \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} + \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} + \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix}$$

$$\therefore S_2 = -1 - 1 - 1 = -3$$

$$\rightarrow S_3 = |A| = 0(0-1) - 1(0-1) + 1(1-0) = 2$$

$\rightarrow$ Put the value of S_1, S_2 and S_3 in equation (1)

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

$$\Rightarrow \lambda^3 - 0\lambda^2 - 3\lambda - 2 = 0 \quad \text{.....} \rightarrow (2)$$

$\rightarrow$ Put $\lambda = -1$ in equation (2)

$$(-1)^3 - 0(-1)^2 - 3(-1) - 2 = -1 - 0 + 3 - 2 = 0$$

$\therefore \lambda = -1$ is one of the root of the equation (2).

$$S_1 = 0, S_2 = -3 \\ \text{and } S_3 = 2$$

Method:5 $\Rightarrow$ Example:2(Continue)

$$\lambda^3 + 0\lambda^2 - 3\lambda - 2 = 0$$

$$\therefore (\lambda + 1)(\lambda^2 - \lambda - 2) = 0$$

$$\therefore (\lambda + 1)(\lambda - 2)(\lambda + 1) = 0$$

$$\therefore (\lambda + 1) = 0 \text{ or } (\lambda - 2) = 0 \text{ or } (\lambda + 1) = 0$$

$$\therefore \lambda = -1, 2, -1 \rightarrow \text{Eigen Values}$$

1	0	-3	-2
1	0	1	2
1	-1	-2	0

$\times +$
 $2 \times -1 = -1$

-2
 -1
 1

Method:5 $\Rightarrow$ Example:2(Continue)

$\Rightarrow$ For $\lambda = -1$, the eigen vector is $(A - \lambda I)X = 0$

$$\Rightarrow (A - (-I))X = 0$$

$$\text{where, } B = A + I = \begin{bmatrix} 0 - (-1) & 1 & 1 \\ 1 & 0 - (-1) & 1 \\ 1 & 1 & 0 - (-1) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } 0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$\Rightarrow \text{Consider, } [B \mid 0] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \end{array} \right]$$

Method:5 $\Rightarrow$ Example:2(Continue)

$\rightarrow$ Consider,
$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \end{array} \right]$$

Applying $R_{12}(-1), R_{13}(-1)$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$x \quad y \quad z$

$\rightarrow$ Let $x = t_1$, & $y = t_2$ & $z = t_3 \in \mathbb{R}$

Applying the Back Substitution, 3

where, $t_1 \neq$ number of variables

Thus, the given system has infinitely many solution: $t_1 - t_2$

Method:5 $\Rightarrow$ Example:2(Continue)

$\Rightarrow$ Thus, the eigen vector corresponding to the eigen value $\lambda = -1$ is

$$\begin{aligned} X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \begin{bmatrix} -t_1 - t_2 \\ t_1 \\ t_2 \end{bmatrix}; t_1, t_2 \in \mathbb{R} \\ &= \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \cdot t_1 + \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \cdot t_2; t_1, t_2 \in \mathbb{R} \end{aligned}$$

$\therefore$ The eigen vectors for $\lambda = -1$ are

Method:5 $\Rightarrow$ Example:2(Continue)

$\Rightarrow$ For $\lambda = 2$, the eigen vector is $(A - \lambda I)X = 0$

$$\Rightarrow (A - 2I)X = 0$$

$$\text{where, } B = A - 2I = \begin{bmatrix} 0 & -2 & 1 & 1 \\ 1 & 0 & -2 & 1 \\ 1 & 1 & 0 & -2 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{bmatrix}$$

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } 0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$\Rightarrow \text{Consider, } [B \mid 0] = \left[\begin{array}{ccc|c} 0 & -2 & 1 & 0 \\ 1 & 0 & -2 & 0 \\ 1 & 1 & 0 & 0 \end{array} \right]$$

Method:5 $\Rightarrow$ Example:2(Continue)

$\Rightarrow$ Consider,

$$\left[\begin{array}{ccc|c} -2 & 1 & 1 & 0 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -2 & 0 \end{array} \right]$$

Applying R_{12}

$$\left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ -2 & 1 & 1 & 0 \\ 1 & 1 & -2 & 0 \end{array} \right]$$

Applying
 $R_{12}(2), R_{13}(-1)$

$$\left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & -3 & 3 & 0 \\ 0 & 3 & -3 & 0 \end{array} \right]$$

Method:5 $\Rightarrow$ Example:2(Continue)

$$\begin{array}{c}
 \text{Applying } R_{12}(2), R_{13}(-1) \\
 \sim \left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & -3 & 3 & 0 \\ 0 & 3 & -3 & 0 \end{array} \right] \\
 \\
 \text{Applying } \left(-\frac{1}{3}\right) R_2 \\
 \sim \left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 3 & -3 & 0 \end{array} \right] \\
 \\
 \text{Applying } R_{23}(-3) \\
 \sim \left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]
 \end{array}$$

Method:5 **Example:2(Continue)**



Applying $R_{23}(-3)$

$$\begin{bmatrix} z & 1 & -1 & 0 \\ y & -2 & 1 & 0 \\ x & 1 & 0 & 0 \end{bmatrix}$$

$\mu \in [\beta, \gamma]$ $\mu \in \mathbb{R}$

$\rho(B) = \rho([B \mid 0])$ Applying the Back Substitution,
where, n = number of variables

$y - z = 0$

$$\Rightarrow y = z$$

$$t \models y \Rightarrow$$

$\Rightarrow x - 2y + z = 0$
 $\Rightarrow x = 2y - z$
 $\Rightarrow x = 2t - t$
 $\Rightarrow x = t$

Method:5 $\Rightarrow$ Example:2(Continue)

$\Rightarrow$ Thus, the eigen vector corresponding to the eigen value $\lambda = 2$ is

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} t \\ t \\ t \end{bmatrix}; t \in \mathbb{R}$$
$$= \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \cdot t; t \in \mathbb{R}$$

$\therefore$ The eigen vector for $\lambda = 2$ is

Method:5 $\Rightarrow$ Example:2(Continue)

$\rightarrow$ The eigen bases for the given matrix A is

$$\left\{ \begin{array}{c} \left[\begin{array}{ccc} -1 & 1 & 0 \end{array} \right]^T, \left[\begin{array}{ccc} -1 & 0 & 1 \end{array} \right]^T, \left[\begin{array}{ccc} 1 & 1 & 1 \end{array} \right]^T \end{array} \right\}$$

or

$$\left\{ \begin{array}{c} \left[\begin{array}{ccc} -1 & 1 & 0 \end{array} \right]^T, \left[\begin{array}{ccc} -1 & 0 & 1 \end{array} \right]^T, \left[\begin{array}{ccc} 1 & 1 & 1 \end{array} \right]^T \end{array} \right\}$$

The eigen vectors for $\lambda = -1$ are

$$\left[\begin{array}{c} -1 \\ 1 \\ 0 \end{array} \right], \left[\begin{array}{c} -1 \\ 0 \\ 1 \end{array} \right],$$

The eigen vector for $\lambda = 2$ is

$$\left[\begin{array}{c} 1 \\ 1 \\ 1 \end{array} \right]^T$$



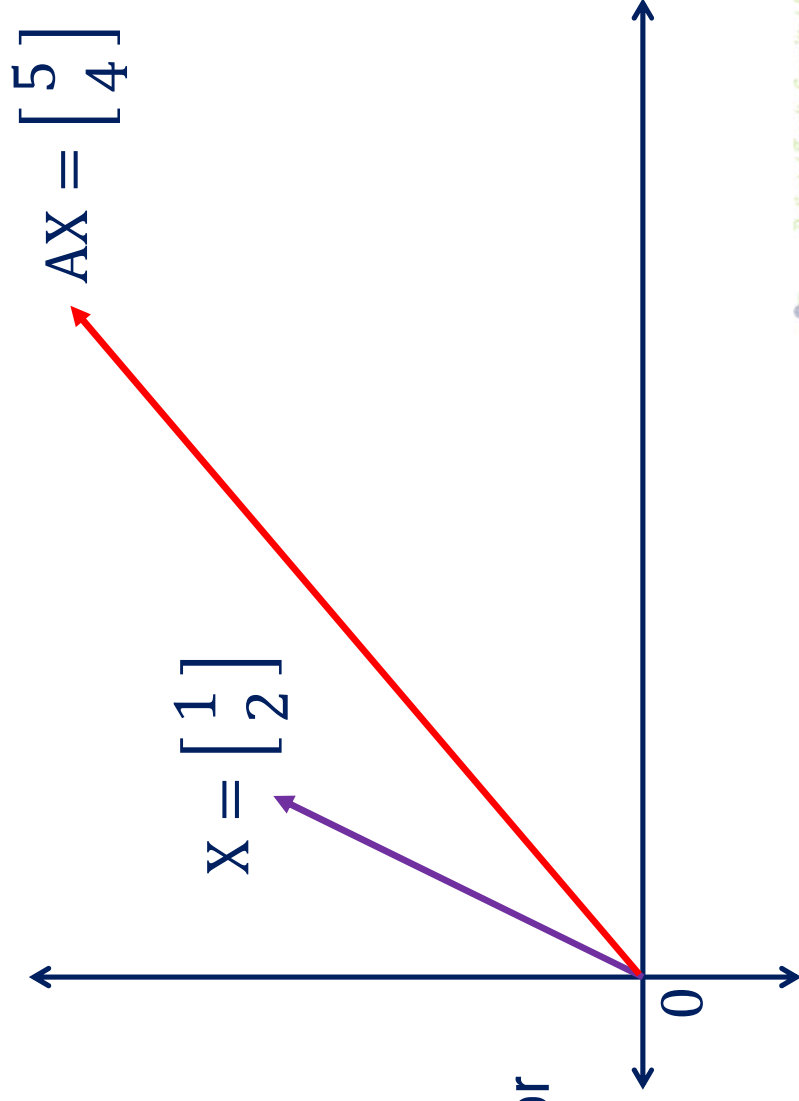
What happens when a matrix is multiplied by a vector?

➡ Let $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ and $X = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

$$\Rightarrow AX = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\Rightarrow AX = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$$

➡ In this case, the resultant vector has been scaled and rotated.



What happens when a matrix is multiplied by a vector?

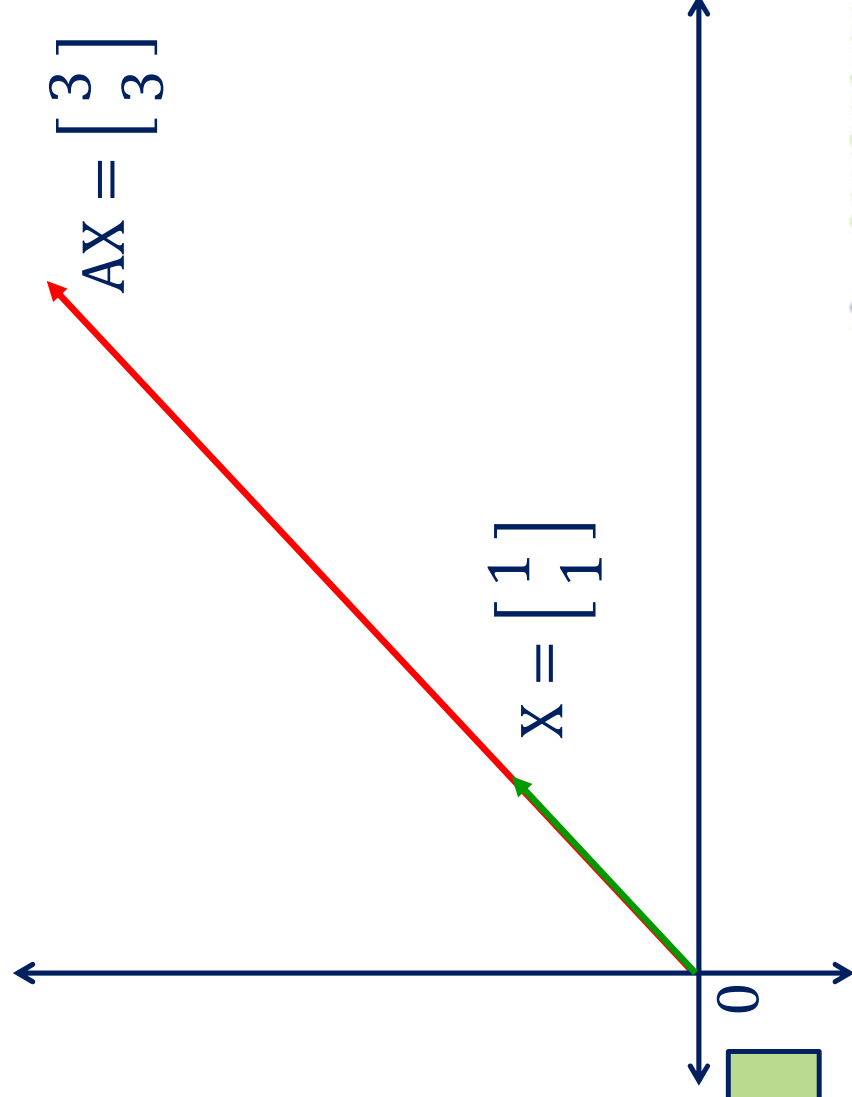
➡ Let $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ and $X = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$$\Rightarrow AX = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\Rightarrow AX = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

$$\Rightarrow AX = 3 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\Rightarrow AX = \lambda X$$



Eigen Vector

Eigen Value

➡ In this case, the resultant vector has scaled but not rotated.

Eigen Value & Eigen Vector

- ➡ Let A be an $n \times n$ matrix, then the scalar λ is called an **Eigen Value** of A if there exist a **non-zero** vector X such that **$AX = \lambda X$** .
- ➡ The non-zero vector X is called an **Eigen Vector** of A corresponding to λ .
- ➡ The set of all eigenvalues of matrix A is called **Spectrum** of A .
- ➡ The solution space of the system **$(A - \lambda I)X = 0$** is called **Eigen Space** of A corresponding to eigen value λ .
i. e. $E_\lambda = \{ X \mid X \text{ is an eigen vector of } A \text{ corresponding to } \lambda \}$.

Notes

- ➔ Eigen value is also known as characteristic root/latent value/proper roots and Eigen Vector is also known as characteristic vector/latent vector/proper vector corresponding to the eigen value λ of the matrix.
- ➔ Sum of Eigen values = Trace of a matrix
- ➔ Product of Eigen values = Determinant of a matrix
- ➔ If one of the eigen values of a matrix A is zero
 $\Rightarrow |A| = 0$
 $\Rightarrow A^{-1}$ does not exist.

Notes(Remember this)

Matrix Form	Name	Eigen Values
$\begin{bmatrix} \mathbf{a} & 0 & 0 \\ 0 & \mathbf{d} & 0 \\ 0 & 0 & \mathbf{f} \end{bmatrix}$	Diagonal Matrix	$\mathbf{a, d, f}$
$\begin{bmatrix} \mathbf{a} & b & c \\ 0 & \mathbf{d} & e \\ 0 & 0 & \mathbf{f} \end{bmatrix}$	Upper Triangular Matrix	$\mathbf{a, d, f}$
$\begin{bmatrix} \mathbf{a} & 0 & 0 \\ b & \mathbf{c} & 0 \\ d & e & \mathbf{f} \end{bmatrix}$	Lower Triangular Matrix	$\mathbf{a, c, f}$

Notes(Remember this)

Matrix Form	Name	Eigen Values
$\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$	Diagonal Matrix	a, b
$\begin{bmatrix} a & b \\ 0 & d \end{bmatrix}$	Upper Triangular Matrix	a, d
$\begin{bmatrix} a & 0 \\ b & c \end{bmatrix}$	Lower Triangular Matrix	a, c

Method:5 → Eigen values and Eigen vectors

Example:3

Find the eigen values of $A = \begin{bmatrix} -5 & 4 & 34 \\ 0 & 0 & 4 \\ 0 & 0 & 4 \end{bmatrix}$. Is A invertible?

Solution:

The given matrix is an upper triangular matrix.

So, the eigen values of $A = -5, 0, 4$



Now, we know that $|A| = \text{Product of eigen values of } A$

$$\Rightarrow |A| = (-5)(0)(4)$$

$$\Rightarrow |A| = 0$$

$$\Rightarrow A^{-1} \text{ does not exist.}$$

So, A is not invertible.

$$\begin{bmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{bmatrix}$$

Upper Triangular Matrix



Method:5 → Eigen values and Eigen vectors

Example:4

Find the eigen values and eigen vector the matrix $A = \begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}$.

Solution:

The given matrix is an upper triangular matrix.

So, the eigen values of $A = 3, 2, 5$

→ For $\lambda = 3$, the eigen vector is $(A - \lambda I)X = 0$

$$\Rightarrow (A - 3I)X = 0$$

$$\text{where, } B = A - 3I = \begin{bmatrix} 0 & 1 & 4 \\ 0 & -1 & 6 \\ 0 & 0 & 2 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } 0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Method:5 $\Rightarrow$ Example:4(Continue)

$\rightarrow$ Consider $[B | 0] = \left[\begin{array}{ccc|ccc} 0 & \boxed{1} & 4 & 0 & 0 & 0 \\ 0 & \boxed{-1} & 6 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 \end{array} \right]$

Applying $R_{12}(1)$

$$\sim \left[\begin{array}{ccc|ccc} 0 & 1 & 4 & 0 & 0 & 0 \\ 0 & 0 & \boxed{10} & 0 & 0 & 0 \\ 0 & 0 & \boxed{2} & 0 & 0 & 0 \end{array} \right]$$

Applying $R_{23}\left(-\frac{1}{5}\right)$

$$\sim \left[\begin{array}{ccc|ccc} 0 & 1 & 4 & 0 & 0 & 0 \\ 0 & 0 & 10 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

Method:5 $\Rightarrow$ Example:4(Continue)

Applying $R_{23} \left(\begin{matrix} 1 \\ -\frac{1}{5} \end{matrix} \right)$ $\sim$

$$\begin{array}{c|ccc} & x & y & z \\ \hline 1 & 0 & 1 & 4 \\ 2 & 0 & 0 & 10 \\ 3 & 0 & 0 & 0 \end{array}$$

$\Rightarrow$ Let, $x = t ; t \in \mathbb{R}$

Applying the Back Substitution,

$\Rightarrow 10z = 0$

$\Rightarrow z = 0$

$\Rightarrow y + 4z = 0$

$\Rightarrow y = -4z$

$\Rightarrow y = 0$

Method:5 $\Rightarrow$ Example:4(Continue)

$\Rightarrow$ Thus, the eigen vector corresponding to the eigen value $\lambda = 3$ is

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} t \\ 0 \\ 0 \end{bmatrix}; t \in \mathbb{R}$$

$$= \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \cdot t; t \in \mathbb{R}$$

$\therefore$ The eigen vector for $\lambda = 3$ is

Method:5 $\Rightarrow$ Example:4(Continue)

$\Rightarrow$ For $\lambda = 2$, the eigen vector is $(A - \lambda I)X = 0$
 $\Rightarrow (A - 2I)X = 0$

$$\text{where, } B = A - 2I = \begin{bmatrix} 11 & 1 & 4 \\ 0 & 0 & 6 \\ 0 & 0 & 3 \end{bmatrix} X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } O = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}$$

$\Rightarrow$ Consider, $[B \mid O] = \left[\begin{array}{ccc|ccc} \dots & \dots & \dots & \dots & \dots & \dots \end{array} \right]$

Convert $[B \mid O]$ into Row-Echelon Form by yourself

Method:5 $\Rightarrow$ Example:4(Continue)

$$\begin{array}{c} x \quad y \quad z \\ \left[\begin{array}{ccc|c} 1 & 1 & 4 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \end{array}$$

$\Rightarrow$ Let, $y = t$; $t \in \mathbb{R}$

Applying the Back Substitution,

$\Rightarrow z = 0$

$\Rightarrow x + y + 4z = 0$

$$\Rightarrow x = -y - 4z$$

$$\Rightarrow x = -t$$

$\Rightarrow$ Thus, the eigen vector corresponding to the eigen value $\lambda = 2$ is

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -t \\ t \\ 0 \end{bmatrix}; t \in \mathbb{R}$$

$$= \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \cdot t; t \in \mathbb{R}$$

$\therefore$ The eigen vector for $\lambda = 2$ is

Method:5 $\Rightarrow$ Example:4(Continue)

$\Rightarrow$ For $\lambda = 5$, the eigen vector is $(A - \lambda I)X = 0$
 $\Rightarrow (A - 5I)X = 0$

$$\text{where, } B = A - 5I = \begin{bmatrix} -2 & 1 & 4 \\ 0 & -3 & 6 \\ 0 & 8 & 0 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } 0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 5 & 5 & 5 \end{bmatrix}$$

$\Rightarrow$ Consider, $[B \mid 0] = \left[\begin{array}{ccc|c} \hline \end{array} \right]$

Convert $[B \mid 0]$ into Row-Echelon Form by yourself



Method:5 $\Rightarrow$ Example:4(Continue)

$$\begin{array}{c} x \quad y \quad z \\ \sim \left[\begin{array}{ccc|c} -2 & 1 & 4 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \end{array}$$

$\Rightarrow$ Let, $z = t ; t \in \mathbb{R}$

Applying the Back Substitution,

$\Rightarrow y - 2z = 0$

$$\Rightarrow y = 2z$$

$$\Rightarrow y = 2t$$

$\Rightarrow -2x + y + 4z = 0$

$$\Rightarrow 2x = y + 4z$$

$$\Rightarrow 2x = 2t + 4t$$

$$\Rightarrow 2x = 6t$$

$$\Rightarrow x = 3t$$

$\Rightarrow$ Thus, the eigen vector corresponding to the eigen value $\lambda = 5$ is

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3t \\ 2t \\ t \end{bmatrix}; t \in \mathbb{R}$$

Method:5 $\Rightarrow$ Example:4(Continue)

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3t \\ 2t \\ t \end{bmatrix}; t \in \mathbb{R}$$

$$= \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} \cdot t; t \in \mathbb{R}$$

$\therefore$ The eigen vector for $\lambda = 5$ is

$\rightarrow$ The Eigen Space for $\lambda = 3$ is

$$E_{\lambda=3} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \cdot t \mid t \in \mathbb{R} \right\}$$

$\rightarrow$ The Eigen Space for $\lambda = 2$ is

$$E_2 = \left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \cdot t \mid t \in \mathbb{R} \right\}$$

$\rightarrow$ The Eigen Space for $\lambda = 5$ is

$$E_5 = \left\{ \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} \cdot t \mid t \in \mathbb{R} \right\}$$

Procedure to find the Eigen Value, Eigen Vector for 2×2 matrix

- 1) Write the Characteristic Equation $\lambda^2 - S_1\lambda + S_2 = 0$ and find S_1, S_2 .
- 2) Find the roots of $\lambda^2 - S_1\lambda + S_2 = 0$, say λ_1, λ_2 .
Eigen values = The solution (λ_1, λ_2) of step:2.
- 3) Solve the system $[A - \lambda_1 I] \cdot \mathbf{X} = \mathbf{0}$ using Gauss Elimination Method.
- 4) Eigen vector = The solution $\mathbf{X}$ of above system.
- 5) Repeat step:4) for λ_2 to find remaining eigen vectors.

Method:5 ➡ Eigen value and Eigen vector

Example:5

Let $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, then find the eigen values, eigen vectors

corresponding to each eigen values of A. Also, write the eigen space for each eigen values.

Solution:

The characteristic equation of the matrix A is

$$\lambda^2 - S_1\lambda + S_2 = 0 \text{} \rightarrow (1)$$

where, $S_1 = \text{trace}(A) = 0$ and $S_2 = |A| = -1$

➡ Put the value of S_1 & S_2 in equation:(1)

$$\Rightarrow \lambda^2 - 0\lambda + (-1) = 0$$

$$\Rightarrow \lambda^2 - 1 = 0$$

$$\Rightarrow \lambda^2 = 1$$

$$\Rightarrow \lambda = \pm 1$$

Method:5 $\Rightarrow$ Example:5(Continue)

$\Rightarrow$ For $\lambda = -1$, the eigen vector is $(A - \lambda I)X = 0$

$$\Rightarrow (A + I)X = 0$$

$$\text{where, } B = A + I = \begin{bmatrix} 0 - (-1) & 1 \\ 1 & 0 - (-1) \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$X = \begin{bmatrix} x \\ y \end{bmatrix} \text{ and } 0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$\Rightarrow$ Consider, $[B \mid 0] = \left[\begin{array}{cc|c} 1 & 1 & 0 \\ 1 & 1 & 0 \end{array} \right]$

Applying $R_{12}(-1)$ $\sim \left[\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & 0 & 0 \end{array} \right]$

Method:5 $\Rightarrow$ Example:5(Continue)

Applying $R_{12}(-1)$

$$\begin{array}{cc|cc} x & y & & \\ \hline 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}$$

$\Rightarrow$ Let, $y = t; t \in \mathbb{R}$

Applying the Back Substitution,

$\Rightarrow x + y = 0$

$\Rightarrow x = -y$

$\Rightarrow x = -t$

$\Rightarrow$ Thus, the eigen vector

corresponding to the eigen value
 $\lambda = -1$ is

$$X = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -t \\ t \end{bmatrix}; t \in \mathbb{R}$$

$$= \begin{bmatrix} -1 \\ 1 \end{bmatrix}; t \in \mathbb{R}$$

$\therefore$ The eigen vector for $\lambda = -1$ is

Method:5 $\Rightarrow$ Example:5(Continue)

$\rightarrow$ For $\lambda = 1$, the eigen vector is $(A - \lambda I)X = 0$

$$\Rightarrow (A - I)X = 0$$

$$\text{where, } B = A - I = \begin{bmatrix} 0 - (1) & 1 \\ 1 & 0 - (1) \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$X = \begin{bmatrix} x \\ y \end{bmatrix} \text{ and } 0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$\rightarrow$ Consider, $[B \mid 0] = \left[\begin{array}{cc|c} -1 & 1 & 0 \\ 1 & -1 & 0 \end{array} \right]$

Applying $R_{12}(1)$ $\sim \left[\begin{array}{cc|c} -1 & 1 & 0 \\ 0 & 0 & 0 \end{array} \right]$

Method:5 $\Rightarrow$ Example:5(Continue)

Applying $R_{12}(1)$

$$\begin{array}{cc|cc} x & y & & \\ -1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}$$

$\Rightarrow$ Let, $y = t; t \in \mathbb{R}$

Applying the Back Substitution,

$\Rightarrow -x + y = 0$

$\Rightarrow x = y$

$\Rightarrow x = t$

$\Rightarrow$ Thus, the eigen vector

corresponding to the eigen value

$\lambda = 1$ is

$$X = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} t \\ t \end{bmatrix}; t \in \mathbb{R}$$

$$= \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot t; t \in \mathbb{R}$$

$\therefore$ The eigen vector for $\lambda = 1$ is

Method:5 $\Rightarrow$ Example:5(Continue)

$\Rightarrow$ The Eigen Space for $\lambda = -1$ is

$$E_{-1} = \left\{ \begin{bmatrix} -1 \\ 1 \end{bmatrix} \cdot t \mid t \in \mathbb{R} \right\}$$

$\Rightarrow$ The Eigen Space for $\lambda = 1$ is

$$E_1 = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot t \mid t \in \mathbb{R} \right\}$$

The Eigen Vector for $\lambda = -1$ is

$$= \begin{bmatrix} -1 \\ 1 \end{bmatrix} \cdot t; t \in \mathbb{R}$$

The Eigen Vector for $\lambda = 1$ is

$$= \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot t; t \in \mathbb{R}$$

Remarks

Matrix	Eigen Values			Matrix	Eigen Values		
A	λ_1	λ_2	λ_3	A	-1	2	5
A^T	λ_1	λ_2	λ_3	A^T	-1	2	5
kA	$k\lambda_1$	$k\lambda_2$	$k\lambda_3$	$4A$	-4	8	20

Remarks

Matrix	Eigen Values			Matrix	Eigen Values		
A	λ_1	λ_2	λ_3	A	-1	2	5
A^{-1}	$\frac{1}{\lambda_1}$	$\frac{1}{\lambda_2}$	$\frac{1}{\lambda_3}$	A^{-1}	-1	$\frac{1}{2}$	$\frac{1}{5}$
A^k	$(\lambda_1)^k$	$(\lambda_2)^k$	$(\lambda_3)^k$	A^3	-1	8	125

Remarks

Matrix	Eigen Values			Matrix	Eigen Values		
A	λ_1	λ_2	λ_3	A	-1	2	5
$A + kI$	$\lambda_1 + k$	$\lambda_2 + k$	$\lambda_3 + k$	$A + 3I$	2	5	8
$A - kI$	$\lambda_1 - k$	$\lambda_2 - k$	$\lambda_3 - k$	$A - 4I$	-5	-2	1

Remarks

Matrix	Eigen Values			Matrix	Eigen Values		
A	λ_1	λ_2	λ_3	A	-1	2	5
A^m	$(\lambda_1)^m$	$(\lambda_2)^m$	$(\lambda_3)^m$	A^3	-1	8	125
$A^m \pm kI$	$\lambda_1^m \pm k$	$\lambda_2^m \pm k$	$\lambda_3^m \pm k$	$A^3 + 4I$	3	12	129

Method:5 Eigen value and Eigen vector

Example:6

Let $A = \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix}$, then find the eigen values of $A, A^{25}, 3A, A^{-1}, A^T$,

$$A + 2I \text{ \& } A^3 - 5I.$$

Solution:

Here, matrix A is a lower triangular matrix.

So, the eigen values of $A = 3, -1$.

Matrix	Eigen Values	Matrix	Eigen Values
A	3	A^T	-1
A^{25}	3^{25}	$A + 2I$	5
$3A$	9	$A^3 - 5I$	22
A^{-1}	$1/3$		-6

Eigen values of A^3

$$= 27, -1$$

Eigen values of $A^3 - 5I$

$$= 27 - 5, -1 - 5$$

$$= 22, -6$$



Method:6 → Algebraic & Geometric Multiplicity



Linearly Dependent and Independent Eigen Vector

Linearly Dependent Vectors

➔ $X_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ & $X_2 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$

Now, $X_2 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$

$$\therefore X_2 = 2 \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\therefore X_2 = 2X_1$$

So, X_1 and X_2 are Linearly Dependent vectors.

Linearly Independent Vectors

➔ $X_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ & $X_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$$\therefore X_2 \neq \alpha X_1 \text{ or } X_1 \neq \alpha X_2$$

So, X_1 and X_2 are Linearly Independent vectors.

For Your Information

➔ Thus, the eigen vector
corresponding to the eigen value
 $\lambda = 1$ is

$$X = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} t \\ t \end{bmatrix}; t \in \mathbb{R}$$

$$= \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot t; t \in \mathbb{R}$$

∴ The eigen vector for $\lambda = 1$ is

L. I. Eigen Vector



Algebraic & Geometric Multiplicity

➡ If Eigen value λ of matrix A is repeated n -times, then n is called the **Algebraic Multiplicity**(A.M.) of λ .

Example: If $\lambda = 1, 1, 2$ then A.M. of $1 = 2$ & A.M. of $2 = 1$

➡ The number of Linearly Independent Eigen vectors corresponding to λ is called **Geometric Multiplicity**(G.M.) of λ .

Example: If for $\lambda = \lambda_1$
the L.I. eigen vectors are

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

then G.M. of $\lambda_1 = 2$

If for $\lambda = \lambda_2$

the L.I. eigen vector is

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix},$$

then G.M. of $\lambda_2 = 1$



Remember this...

- ➔ G.M. of $\lambda = n - \rho(A - \lambda I)$; where, n = number of variables
- ➔ G.M. of $\lambda \leq$ A.M. of λ

Procedure to find A. M. and G. M. of Eigen Values

- ➔ Find the eigen values of the given matrix.
- ➔ Write A.M. of each eigen values.
- ➔ Find eigen vectors(if asked) for each eigen value.
- ➔ Write G.M., which is number of L.I. eigen vectors.
- ➔ Otherwise use the following formula for G.M.
G.M. of $\lambda = n - \rho(A - \lambda I)$; where, n = number of variables



Method:6 Algebraic & Geometric Multiplicity

Example:1

Find the eigen values and corresponding eigen vectors of the matrix

$$A = \begin{bmatrix} -3 & -7 & -5 \\ 2 & 4 & 3 \\ 1 & 2 & 2 \end{bmatrix}. \text{ Also, find the algebraic and geometric multiplicity for each eigen value.}$$

Solution:

The characteristic equation of the matrix A is given by

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0 \text{} \rightarrow (1)$$

where, $S_1 = \text{trace}(A)$

$$= -3 + 4 + 2 = 3$$

 $S_2 = \text{Sum of minors of principal diagonal elements}$

$$\therefore S_2 = \begin{vmatrix} 4 & 3 \\ 2 & 2 \end{vmatrix} + \begin{vmatrix} -3 & -5 \\ 1 & 2 \end{vmatrix} + \begin{vmatrix} -7 & -5 \\ 2 & 4 \end{vmatrix}$$



Method:6 $\Rightarrow$ Example:1(Continue)

$$\therefore S_2 = \begin{vmatrix} 4 & 3 \\ 2 & 2 \end{vmatrix} + \begin{vmatrix} -3 & -5 \\ 1 & 2 \end{vmatrix} + \begin{vmatrix} -3 & -7 \\ 2 & 4 \end{vmatrix}$$

$$\therefore S_2 = 2 - 1 + 2 = 3$$

$$\Rightarrow S_3 = |A| = -3(8 - 6) + 7(4 - 3) - 5(4 - 4) = 1$$

$\Rightarrow$ Put the value of S_1, S_2 and S_3 in equation (1)

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

$$\Rightarrow \lambda^3 - 3\lambda^2 + 3\lambda - 1 = 0$$

$$\Rightarrow (\lambda - 1)^3 = 0$$

$$\Rightarrow \lambda = 1, 1, 1$$

$$\therefore \text{A.M. of } 1 = 3$$

$$S_1 = 3, S_2 = 3 \\ \text{and } S_3 = 1$$

$$(a - b)^3 = a^3 - b^3 - 3ab(a - b)$$

Method:6 $\Rightarrow$ Example:1(Continue)

$\Rightarrow$ For $\lambda = 1$, the eigen vector is $(A - \lambda I)X = 0$

$$\Rightarrow (A - I)X = 0$$

$$\text{where, } B = A - I = \begin{bmatrix} -3 & -1 & -7 & -5 \\ 2 & 4 & -1 & 3 \\ 1 & 2 & 2 & -1 \end{bmatrix} = \begin{bmatrix} -4 & -7 & -55 \\ 2 & 3 & 33 \\ 1 & 2 & 11 \end{bmatrix}$$

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } 0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$A = \begin{bmatrix} -3 & -7 & -5 \\ 2 & 4 & 3 \\ 1 & 2 & 2 \end{bmatrix}$$

$\Rightarrow$ Consider, $[B \mid 0] = \left[\begin{array}{ccc|c} \end{array} \right]$

Method:6 $\Rightarrow$ Example:1(Continue)

$\Rightarrow$ Consider, $[B | 0] =$

$$\left[\begin{array}{ccc|c} -4 & -7 & -5 & 0 \\ 2 & 3 & 3 & 0 \\ 1 & 2 & 1 & 0 \end{array} \right]$$

Applying R_{13}

$$\sim \left[\begin{array}{ccc|c} \boxed{} & 3 & 3 & 0 \\ \boxed{2} & 3 & 3 & 0 \\ \boxed{} & 3 & 3 & 0 \end{array} \right]$$

Applying $R_{12}(-2), R_{13}(4)$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 1 & -1 & 0 \end{array} \right]$$

Method:6 $\Rightarrow$ Example:1(Continue)

Applying
 $R_{12}(-2), R_{13}(4)$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 1 & -1 & 0 \end{array} \right]$$

Applying $R_{23}(1)$

$$\sim \left[\begin{array}{ccc|c} x & y & z & \\ 1 & 2 & 1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$\Rightarrow$ Let, $z = t; t \in \mathbb{R}$

Applying the Back Substitution,

$\Rightarrow -y + z = 0$

$$\Rightarrow y = z$$

$$\Rightarrow y = t$$

$\Rightarrow x + 2y + z = 0$

$$\Rightarrow x = -2y - z$$

$$\Rightarrow x = -2t - t$$

$$\Rightarrow x = -3t$$

Method:6 $\Rightarrow$ Example:1(Continue)

$\Rightarrow$ Thus, the eigen vector corresponding to the eigen value $\lambda = 1$ is

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -3t \\ t \\ t \end{bmatrix}; t \in \mathbb{R}$$

$$= \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix} \cdot t; t \in \mathbb{R}$$

$\therefore$ The L.I. eigen vector for $\lambda = 1$ is

$\therefore$ G.M. of 1 = 1

Method:6 Algebraic & Geometric Multiplicity

Example:2

Find the A.M. and G.M. for each eigen values of $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$.

Solution:

The characteristic equation of the matrix A is given by

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0 \text{} \rightarrow (1)$$

where, $S_1 = \text{trace}(A)$

$$= 0 + 0 + 0 = 0$$

 $S_2 = \text{Sum of minors of principal diagonal elements}$

$$\therefore S_2 = \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} + \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} + \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix}$$

Method:6 $\Rightarrow$ Example:2(Continue)

$$\therefore S_2 = \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} + \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} + \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix}$$

$$\therefore S_2 = -1 - 1 - 1 = -3$$

$$\rightarrow S_3 = |A| = 0(0-1) - 1(0-1) + 1(1-0) = 2$$

$\rightarrow$ Put the value of S_1, S_2 and S_3 in equation (1)

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

$$\Rightarrow \lambda^3 - 0\lambda^2 - 3\lambda - 2 = 0 \quad \text{.....} \rightarrow (2)$$

$\rightarrow$ Put $\lambda = -1$ in equation (2)

$$(-1)^3 - 0(-1)^2 - 3(-1) - 2 = -1 - 0 + 3 - 2 = 0$$

$\therefore \lambda = -1$ is one of the root of the equation (2).

$$S_1 = 0, S_2 = -3 \\ \text{and } S_3 = 2$$

Method:6 $\Rightarrow$ Example:2(Continue)

$$\lambda^3 + 0\lambda^2 - 3\lambda - 2 = 0$$

$$\therefore (\lambda + 1)(\lambda^2 - \lambda - 2) = 0$$

$$\therefore (\lambda + 1)(\lambda - 2)(\lambda + 1) = 0$$

$$\therefore (\lambda + 1) = 0 \text{ or } (\lambda - 2) = 0 \text{ or } (\lambda + 1) = 0$$

$$\therefore \lambda = -1, 2, -1$$

So, A.M. of $-1 = 2$ & A.M. of $2 = 1$

$$\begin{vmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{vmatrix} = 0$$

$$\lambda \begin{vmatrix} \lambda & 0 \\ 0 & \lambda \end{vmatrix} - 0 + 0 = 0$$

$$\lambda^2 - 3\lambda - 2 = 0$$

$$\lambda^2 - \lambda - 2 = 0$$

$$(\lambda + 1)(\lambda - 2) = 0$$

$$\lambda = -1, 2, -1$$

Method:6 $\Rightarrow$ Example:2(Continue)

$\rightarrow$ G.M. of $\lambda = n - \rho(A - \lambda I)$

where, n = number of variables

G.M. of $(-1) = n - \rho(A + I) \cdots \cdots \cdots (3)$

$$\rightarrow A + I = \begin{bmatrix} 0 - (-1) & 1 & 1 \\ 1 & 0 - (-1) & 1 \\ 1 & 1 & 0 - (-1) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$\begin{array}{c} x \quad y \quad z \\ \left[\begin{array}{ccc} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array} \right] \end{array}$$

From equation (3)

$$\begin{aligned} \text{G.M. of } (-1) &= n - \rho(A + I) \\ &= 3 - 1 \\ &= 2 \end{aligned}$$

Applying
 $R_{12}(-1), R_{13}(-1)$

Method:6 $\Rightarrow$ Example:2(Continue)

$\rightarrow$ G.M. of $\lambda = n - \rho(A - \lambda I)$

where, n = number of variables

G.M. of 2 = $n - \rho(A - 2I)$ $\cdots \cdots \cdots (4)$

$$\rightarrow A - 2I = \begin{bmatrix} 0 & -2 & 1 & 1 \\ 1 & 0 & -2 & 1 \\ 1 & 1 & 0 & -2 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{bmatrix} \quad A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

Applying R_{12} $\sim$ $\begin{bmatrix} 1 & -2 & 1 \\ -2 & 1 & 1 \\ 1 & 1 & -2 \end{bmatrix}$

Method:6 $\Rightarrow$ Example:2(Continue)

Applying R_{12}

$$\sim \begin{bmatrix} \boxed{1} & -2 & 1 \\ \boxed{-2} & 1 & 1 \\ \boxed{1} & 1 & -2 \end{bmatrix}$$

Applying $R_{12}(2), R_{13}(-1)$

$$\sim \begin{bmatrix} 1 & -2 & 1 \\ 0 & -3 & 3 \\ 0 & 3 & -3 \end{bmatrix}$$

Method:6 $\Rightarrow$ Example:2(Continue)

Applying
 $R_{12}(2), R_{13}(-1)$

$$\sim \begin{bmatrix} 1 & -2 & 1 \\ 0 & -3 & 3 \\ 0 & 3 & -3 \end{bmatrix}$$

Applying $R_{23}(1)$

$$\sim \begin{bmatrix} 1 & -2 & 1 \\ 0 & -3 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

From equation (3)

$$\text{G.M. of 2} = n - \rho(A - 2I)$$

$$\text{G.M. of 2} = n - \rho(A - 2I)$$

$$= 3 - 2 = 1$$



Mathematics-I
GTU#(3110014)

**Thank
You**



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$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \dots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix}$$

Matrix

